



Embedded and IoT (45 days)

Recap of Embedded C

- Datatypes
- Array
- Conditional Statements
- Functions / Callback function
- Structures
- Pointers
- Storage classes

EMBEDDED SYSTEM

- Basic Concepts
- Sensors/Actuators
- Microcontroller units and Architecture (Arduino)
- Interfaces (serial port, SPI, I2C, UART)
- ADC, Timers, RTC, Interrupts, Polling, DMA
- Memory architecture and handling (stack, heap, cache)
- Application driven Selection of Microcontrollers

SIGNAL AND ACQUISITION

- Analog and Digital
- Industrial Terminology

INTRODUCTION TO IOT

- A What is IoT
- IoT application in different domains
- Trends in IoT Market
- Smart things
- Gateways
- Middleware
- Edge vs Cloud functional partitioning



IOT ARCHITECTURE

- Architecture.
- Tech Stack.
- Hardware Development Platforms
- Software Development Platforms
- Communication Protocols
- Power Requirements in IoT
- Cloud, its components and IoT
- Data Streaming and IoT
- Data Store and IoT
- Analytics & Visualization and IoT
- IoT Security

LIVE USE CASES

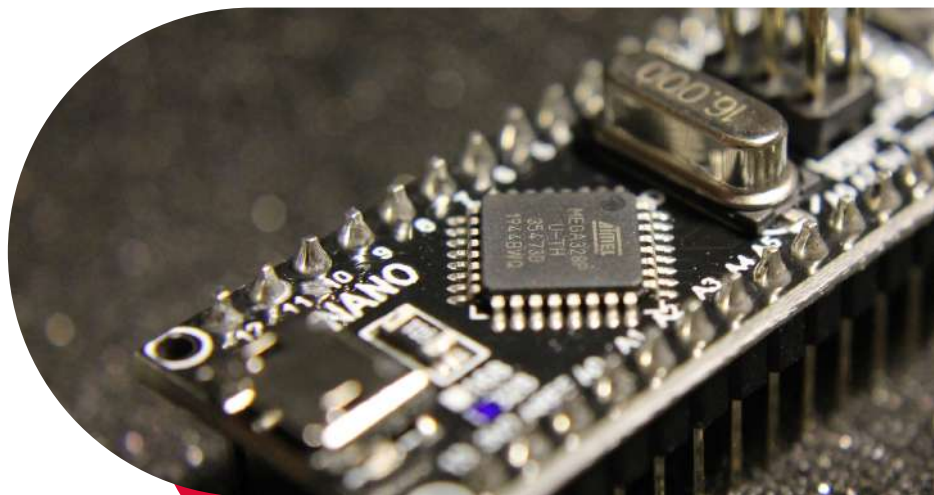
- Claims to cold chain
(Haldiram chain: Customers complaining to not temperature of food items)
- Automation of irrigation
- H/W Components
Client, Device, Rule, Alarm, Device Data, Event
Sensor data being sent to Cloud platform
Actuation through BLE
Visualization through real time dashboard
How the problem gets solved through big data and visualization

NODEMCU (ESP8266)

- Install IDE Software
- Introduction to NODEMCU(Esp8266)
- Pinout of NODEMCU
- Programming Of NODEMCU
- Weather monitoring solution
- Automatic street lights

IOT PROTOCOLS

- HTTP/REST
- MQTT
- COAP
- AMQP
- WEBSOCKETS
- 6LoWPAN



NODE-RED

- Installation of Nodejs,
- Installation of Node-Red,
- Building your first flows
- Basic nodes and flows
- A tour of the core nodes
- The Node-RED programming model
- Dashboards and UI techniques
- Local broker installation,
- Connection between local broker and Node-RED

CLOUD COMPUTING

- Cloud Computing
- Benefits of Cloud
- Deployment Models
- AWS-IOT Core/ SageMaker
- Greengrass
- Google IOT Cloud
- Thingspeak
- IBM Bluemix
- AWS
- AZURE
- Google Firebase
- SQL and NoSQL Data bases
- Integration with MySQL
- MongoDB
- IoT Cloud Platform: Device Management

EDGE AI

- Standards and best practices
- Types of analytics and machine learning
- Supervised/ Un-supervised learning
- Python for IoT analytics
- Computer vision for image data
- Anomaly detection
- Edge Impulse/ TinyML

INDUSTRY4.0

- Introduction to Industry 4.0
- Road to Industry 4.0
- Role of data, information, knowledge and collaboration in future organizations.
- Related Disciplines, System, Technologies for enabling Industry 4.0
- IoT Deployment with legacy systems PLC/SCADA/OPC-UA



Embedded & IoT Developer 3 months

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COMMUNICATION

- Introduction to communication architecture - Network protocol stack
- Different protocols
- RF: ZigBee, BlueTooth, BLE, Zwave, Google thread, Mesh network.
- Communication Channels: GSM/GPRS, 2G, 3G, 4G, LTE, WiFi, PLC
- LPWAN - LoRa & LoRaWAN, Sigfox, Weightless. 3GPP - NB-IoT, LTE -M
- Comparison between different RF Technologies.
- IPv4 addressing problem for IOT and introduction to IPV6

ARM MICROCONTROLLER (STM32)

- Introduction to Microcontrollers (Based on Architecture Selected)
- Instruction set architecture - pipelines
- Registers banks/mode and states
- Clock and reset systems, timers and Memory Organisation
- IDE Configuration, Linker Script, Compiler Optimization Options
- Exception/Interrupt handling
- C and Embedded C Migration
- Memory /IO and peripherals - (I2C/SPI/UART/GPIO/Timer/PWM/ADC)
- Debugging tools like JTAG debugger, CRO
- MMU
- CubeMX

IOT PROTOCOLS

- HTTP/REST
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- IoT Cloud Platform: Device Management

IOT SECURITY

- Introduction to mbed TLS and SSL
- Importance of IoT application
- AES encryption

IOT CLOUD SERVICES

- IoT Cloud Architecture
- Services-SAAS-PAAS-IAAS

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Adv Embedded & IoT System 6 months

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PYTHON

- Data types and type conversion
- Variables and basic operations
- Flow control
- Loops
- Lists, Set and Tuple
- Dictionary
- Functions
- File Handling
- Class and Objects
- Error/ Exception Handling

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RPI

- Introduction to Raspberry pi
- Setup and installation
- Raspberry pi OS walkthrough
- Network setup
- Code execution and data generation
- Sending data to cloud

ARM PROCESSOR AND ARM BASED SUBSYSTEM

- ARM Processor Architecture
- Memory /IO and peripherals - (I2C/SPI/UART/GPIO/Timer/PWM/ADC)
- Debugging tools like JTAG debugger, CRO
- MMU
- ARM Based Embedded Subsystem Architecture
- Multicore Architecture

EMBEDDED LINUX

LINUX INTERNALS

- Linux internals
- Linux introduction and installation.
- Linux Shell Commands
- Shell Scripting
- C Programming in Linux
- Make Files
- Process Management
- File Operation
- Signals in Linux
- Linux Scheduler & Memory Management
- Linux Multi-Threading Programming
- Inter Process Communication (Pipes, Semaphores, Queues, Shared Memory)
- Network Programming in Linux

SOCKETS

- Socket connection
- Socket Attributes
- Creating a Socket
- Socket Addresses
- Naming a Socket
- Creating a Socket Queue
- Accepting Connections
- Requesting Connections
- Closing a Socket
- Socket Communications Processes and Signals
- What is process?
- Process Structure
- The Process Table
- Viewing Processes
- System Processes



MODULE-II: ELINUX PORTING

- Introduction, Setup & Hardware
- Toolchain & Hardware Practical's
- Bootloader U-Boot
- U-Boot Porting
- Customizing Bootloader
- Linux Kernel
- Kernel Porting & Compilation
- Kernel Modification
- Root File System
- Embedded Application Development

MODULE-III: LINUX - DEVICE DRIVERS

- Introduction and Arch of Linux Device Drivers
- Kernel Module Programming
- Character Device Drivers
- Interrupts in Device Driver
- Interrupt Handling & Bottom Half
- Process creation calls (fork,vfork,execv)
- LDDM (Linux Device Driver Model)
- Writing and testing of Board File
- Device Tree
- Advance Device Drivers
- Debugging Techniques using GDB & valgrind other tools.

INTER-PROCESS COMMUNICATION

- Pipes
- Semaphores, message queues and shared memory
- Shared Memory
- Message Queues

MODULE-IV: YOCTO

- Yocto Architecture
- Recipes defines everything in Yocto
- Layers makes Yocto Modular & Structured
- Adding new Hardware support using BSP Layers
- Custom Distribution & Images
- Creating SDK using Yocto for Application Development

IOT PROTOCOLS

- Yocto HTTP/REST, MQTT, COAP, AMQP, WEBSOCKETS, 6LoWPAN

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USECASES

- Automobile
- Electrical Vehicle

